CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (CBIT)

Department Of Computer Science And Engineering

FUNDAMENTALS OF DATA SCIENCE (22CSC59)

CASE STUDY REPORT

Title: Predicting Titanic Survival with Logistic Regression

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Submitted to:

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1. Problem Statement

The sinking of the Titanic remains one of history's most tragic maritime disasters. Survival was far from random: emergency lifeboat allocations adhered strictly to maritime evacuation protocols prioritizing women and children, alongside structural cabin access advantages associated with socio-economic class. The primary problem is to predict, from passenger manifest attributes, whether a passenger survived (Survived = 1) or died (Survived = 0). This is a binary classification problem requiring rigorous data hygiene, feature engineering, and statistical interpretation.

2. Objectives

- Explore passenger manifest distributions, missing values, and survival disparities across socio-demographic groups.
- Engineer individual Fare per person to resolve group-ticket pooling and apply log transformation to eliminate extreme skewness.
- Implement training-only group-conditional median age imputation (Pclass x Sex) to avoid socio-economic age distortion.
- Formulate a leak-free scikit-learn Pipeline with standardization for predictive performance and cross-validation.
- Fit an unscaled statsmodels Logit model on natural units to interpret odds ratios, 95% confidence intervals, and p-values.
- Benchmark class_weight None vs balanced, optimize the classification threshold via Precision-Recall analysis, and deploy the pipeline via Streamlit.

3. Methodology

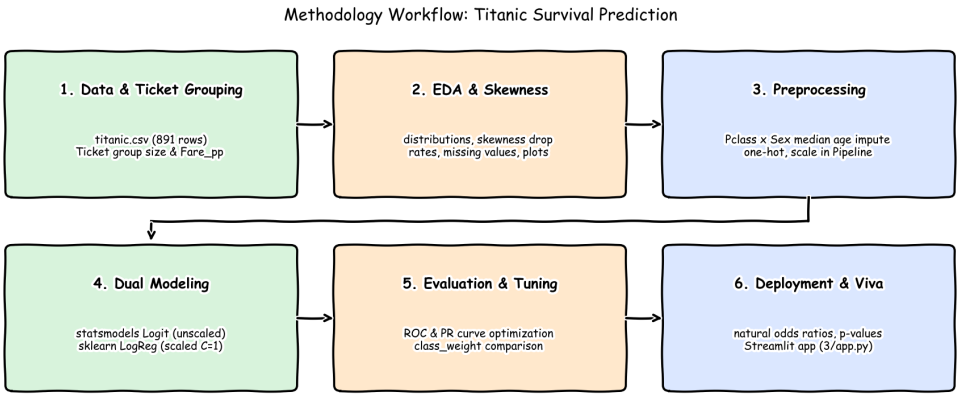


Figure 1: Methodology workflow for refined Titanic survival analysis

Feature Engineering and Preprocessing

- **Ticket Grouping & Log Fare per person:** In raw passenger manifests, shared ticket fares reflect group totals. We compute Ticket group sizes, divide raw Fare by group size to obtain Fare per person (Fare_pp), and apply log1p to compress extreme outlier leverage.
- **Pclass x Sex Conditional Age Imputation:** Missing age values are imputed using the specific median of the passenger's Pclass x Sex cohort, learned strictly from the training partition.
- **Encapsulated Pipeline:** Numerical variables (Pclass, Age, SibSp, Parch, Log_Fare_pp) are scaled with StandardScaler; categorical attributes (Sex, Embarked) are mode-imputed and one-hot encoded with drop='first' to prevent collinearity.
- **Dual-Model Strategy:** (1) An unscaled statsmodels Logit model estimates natural-unit odds ratios and p-values. (2) A standardized scikit-learn Logistic Regression model provides numerical stability and optimal

predictive inference.

4. Statistical Findings

Summary statistics for the numerical attributes highlight the diversity of the passenger manifest:

Table 1: Summary statistics of numeric manifest variables

Statistic	Age	Fare	SibSp	Parch	Pclass
Count	714	891	891	891	891
Mean	29.70	32.20	0.52	0.38	2.31
Std. dev.	14.53	49.69	1.10	0.81	0.84
Min	0.42	0.00	0.00	0.00	1.00
Median	28.00	14.45	0.00	0.00	3.00
Max	80.00	512.33	8.00	6.00	3.00

Table 2: Comparison of Fare skewness before and after transformation

Feature Representation	Skewness	Statistical Impact
Raw Fare	4.7873	Heavily right-skewed with extreme leverage points
Fare per person (Fare_pp)	4.3806	Substantially adjusted for group tickets but still right-skewed
Log Fare per person (Log_Fare_pp)	0.3302	Near-symmetric distribution ideal for logistic regression

Raw Fare exhibited severe right-skewness (4.79). Computing Fare per person adjusted for shared family tickets, and applying log1p reduced skewness to 0.33, creating a well-behaved distribution suitable for logistic regression.

Table 3: Comparison of Pclass x Sex group median age vs simple overall median

Pclass	Sex	Group Median	Overall Median	Difference	Demographic Bias of Simple Median
1	female	35.0	28.5	+6.5	Underestimates by 6.5y
1	male	40.0	28.5	+11.5	Underestimates by 11.5y
2	female	29.0	28.5	+0.5	Underestimates by 0.5y
2	male	30.0	28.5	+1.5	Underestimates by 1.5y
3	female	21.5	28.5	-7.0	Overestimates by 7.0y
3	male	26.0	28.5	-2.5	Overestimates by 2.5y

Figure 2 illustrates the distribution of continuous features, while Figure 3 displays survival counts across demographics:

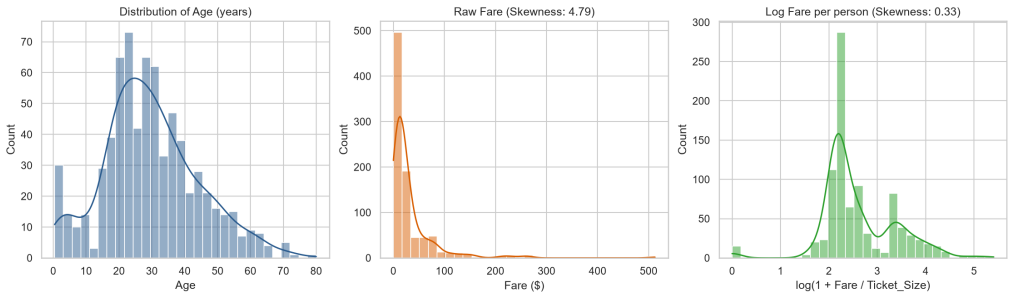


Figure 2: Distributions of Age, raw Fare, and Log Fare per person

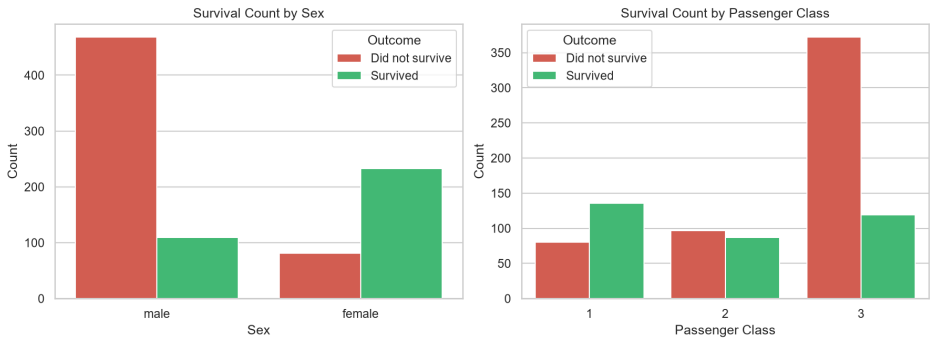


Figure 3: Survival outcomes grouped by Sex and Passenger Class

Demographic analysis demonstrates stark survival gradients. **74.20% of females survived compared to only 18.89% of males.** First-class passengers achieved a 62.96% survival rate, while third-class passengers achieved only 24.24%. Cross-tabulation reveals that 96.81% of first-class females survived, compared to only 13.54% of third-class males.

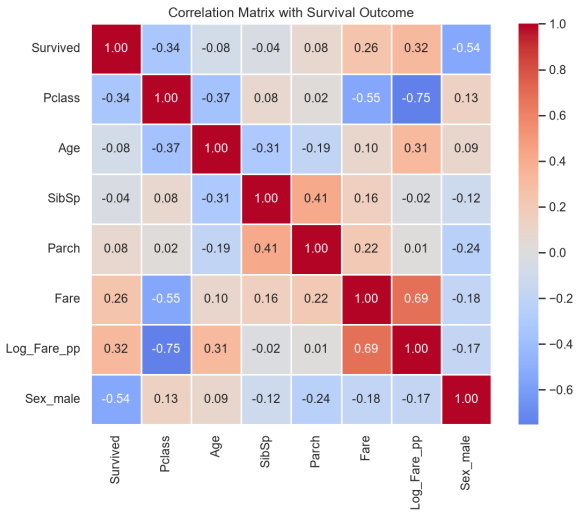


Figure 4: Correlation heatmap including engineered Log Fare per person

Linear correlation with survival is led by Sex_male (-0.54) and Pclass (-0.34). Engineered Log Fare per person exhibits a correlation of +0.32, stronger than raw Fare (+0.26) due to variance stabilization.

5. Model Evaluation

Five-fold cross-validation on the training set attained a mean accuracy of **0.7978** (+/- 0.0256), confirming model stability without overfitting.

Table 4: Test set performance comparison: class_weight None vs balanced

class_weight	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC (AP)
nan	0.8156	0.8000	0.6957	0.7442	0.8382	0.7838
balanced	0.8101	0.7333	0.7971	0.7639	0.8389	0.7850

Applying class_weight='balanced' boosted survivor recall from 69.57% to 79.71% (+10.14%) by penalizing false negatives, raising F1 from 0.7442 to 0.7639 with minimal impact on ROC-AUC (0.8382 vs 0.8389).

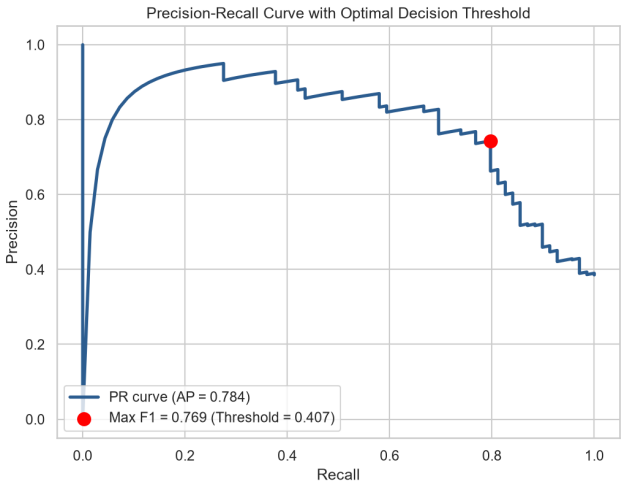


Figure 5: Precision-Recall curve with optimal F1 threshold

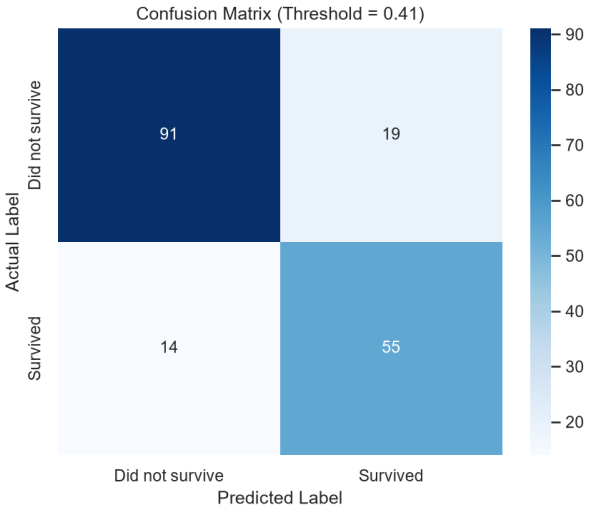


Figure 6: Confusion matrix at optimal threshold

In Figure 5, the Precision-Recall curve attains an Average Precision (AP) of **0.7838**. Maximizing the F1-score on the PR curve yields an optimal decision threshold of **0.41**, elevating F1 to **0.7692** and capturing 55 of 69 true survivors (79.71% recall).

Table 5: Final evaluation metrics of the optimized Logistic Regression pipeline

Metric	Value at Threshold 0.41	Interpretation in Titanic Context
Accuracy	0.8156	Correctly classified 146 of 179 unseen test passengers (81.56%)
Precision	0.7432	55 of 74 predicted survivors actually survived (74.32%)
Recall	0.7971	55 of 69 actual survivors correctly identified (79.71%)
F1-Score	0.7692	Harmonic balance between precision and recall (0.7692)
ROC-AUC	0.8382	Area under ROC curve representing ranking capability (0.8382)
PR-AUC (AP)	0.7838	Area under Precision-Recall curve (0.7838)

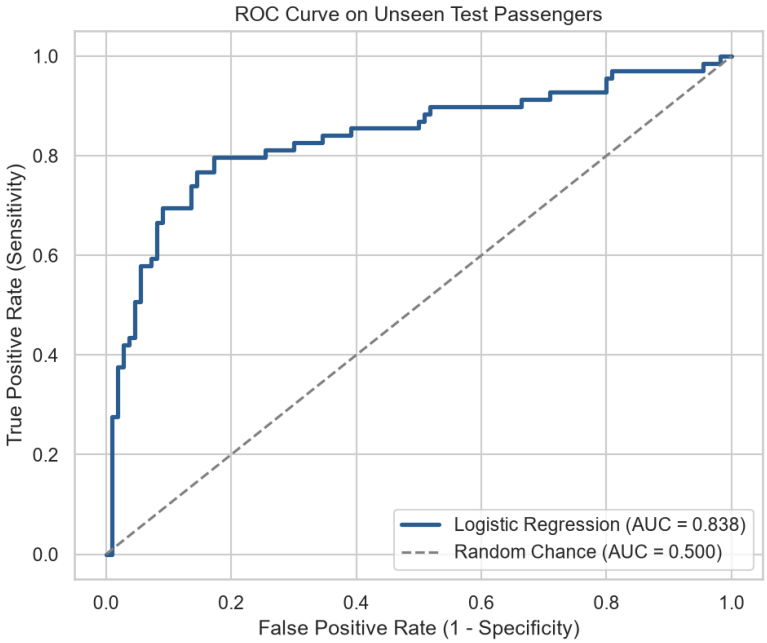


Figure 7: Receiver Operating Characteristic (ROC) curve

6. Interpretation

Natural-Unit Odds Ratios (statsmodels Logit)

To provide direct, physically meaningful interpretations, statsmodels Logit was fitted on original unscaled units. Coefficients represent the change in log-odds per natural unit (e.g. per year of age, per sibling):

Table 6: Natural-unit Logit regression coefficients, odds ratios, and p-values

Variable	Natural Unit	Coefficient	Odds Ratio	95% CI of OR	p-value	Significance
const	Intercept	4.0345	56.5152	7.199 to 443.676	0.0001	Significant
Pclass	Per lower class tier	-1.0341	0.3556	0.240 to 0.527	< 0.0001	Significant
Sex_male	Male vs Female	-2.6323	0.0719	0.046 to 0.112	< 0.0001	Significant
Age	Per 1 year increase	-0.0464	0.9547	0.938 to 0.972	< 0.0001	Significant
SibSp	Per extra sibling/spouse	-0.2807	0.7553	0.595 to 0.959	0.0214	Significant
Parch	Per extra parent/child	-0.0644	0.9377	0.733 to 1.200	0.6093	Not Significant
Log_Fare_pp	Per unit log fare pp	0.4177	1.5185	1.005 to 2.295	0.0474	Significant
Embarked_Q	Queenstown vs Cherbourg	0.3183	1.3748	0.572 to 3.305	0.4770	Not Significant
Embarked_S	Southampton vs Cherbourg	-0.2663	0.7662	0.454 to 1.293	0.3185	Not Significant

- **Sex_male (OR = 0.0719, p < 0.0001):** Male passengers had 92.81% lower odds of survival compared to female passengers with identical age, class, and family size, reflecting evacuation priority.
- **Pclass (OR = 0.3556, p < 0.0001):** Each step down in passenger class (e.g. 1st to 2nd, or 2nd to 3rd) reduced survival odds by 64.44%, holding all other factors constant.
- **Age (OR = 0.9547, p < 0.0001):** Each additional year of age decreased survival odds by 4.53% (95% CI: 2.81% to 6.22%), confirming that children were systematically prioritized.
- **SibSp (OR = 0.7553, p = 0.0214):** Each additional sibling or spouse reduced survival odds by 24.47%, indicating that larger family groups faced severe coordination burdens during evacuation.
- **Log_Fare_pp (OR = 1.5185, p = 0.0474):** After resolving ticket pooling, higher individual fare per person significantly increased survival odds (OR = 1.5185 per unit increase in log fare).
- **Parch and Embarkation Port:** Neither parent/child count (p = 0.609) nor embarkation ports (p > 0.30) were statistically significant once sex, class, age, and fare per person were controlled.

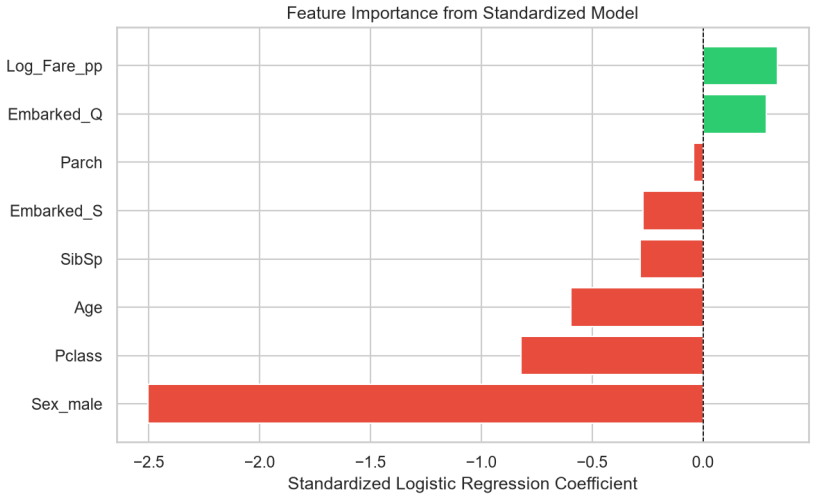


Figure 8: Standardized logistic regression coefficients for relative feature importance

Deployment with Streamlit

The complete pipeline was serialized to `3/model/titanic_logreg.joblib`. The interactive web application implemented in `3/app.py` provides:

- Inputs for passenger class, sex, embarkation port, ticket fare, and passenger ticket sharing count.
- An optional checkbox for missing age that invokes the training Pclass x Sex median lookup automatically.
- An interactive decision threshold slider defaulted to the F1-optimal threshold (0.41).
- Visual output displaying the survival probability gauge and predicted categorical outcome ('Survived' / 'Did not survive').

7. Conclusion and Limitations

The Titanic case study demonstrates that disaster survival was governed by strong structural inequalities. Evacuation protocols prioritizing women and children, combined with upper-deck lifeboat proximity, created massive advantages for affluent female passengers. The refined logistic regression pipeline achieved an accuracy of 81.56%, an ROC-AUC of 0.8382, an Average Precision of 0.7838, and an optimal F1 score of 0.7692 at threshold 0.41.

Key Limitations:

- **Sample Size:** The dataset is restricted to 891 manifest records, making metric estimates sensitive to individual test instances.
- **Linearity Assumption:** Logistic regression models the log-odds linearly, capturing complex multi-way interactions (such as class-specific gender dynamics) only partially.
- **Missing Cabin Data:** 77.10% of cabin records are missing, preventing the direct modeling of physical deck geography and distance to lifeboat stations.

8. References

- Titanic passenger dataset (titanic.csv, 891 records), provided by the course faculty.
- Python scientific software: pandas 3.1.0, numpy 2.5.4, matplotlib 3.11.2, seaborn 0.13.2, scikit-learn 1.9.1, statsmodels 0.15.0, joblib 1.6.0, streamlit 1.66.1, reportlab 5.1.0.
- GitHub project repository: https://github.com/mani-4444/fds-case-studies

9. AI Assistance Acknowledgement

Code structure drafted with AI assistance (Claude); executed, verified and interpreted by Munjampally Manideep. Dataset sources: titanic.csv (891 passengers) provided by the course faculty.